

REVIEW

The gut microbiome in pediatric patients undergoing allogeneic hematopoietic stem cell transplantation

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Abstract

The gut microbiome (GM) has been associated with different clinical outcomes in the context of allogeneic hematopoietic stem cell transplantation (HSCT). Large multicenter cohort studies in adults have found significant correlations with overall survival, relapse, and incidence of complications. Moreover, GM is already a promising target for therapeutic interventions. However, few data are available in children, a population presenting unique features and challenges. During childhood, the GM evolves rapidly with large structural fluctuations, alongside with the maturation of the immune system. Furthermore, the HSCT procedure presents significant differences in children. These considerations underline the importance of a specific focus on the pediatric setting, and the role of GM and its age-dependent trajectory in influencing the immunity reconstitution and clinical outcomes. This review provides a comprehensive overview of the available evidence in the field of GM and pediatric HSCT, highlighting age-specific issues and discussing GM-based therapeutic approaches.

KEYWORDS

bloodstream infections, fecal microbiota transplantation, graft-versus-host disease, gut microbiome, hematopoietic stem cell transplantation, pediatric

1 | INTRODUCTION

Recent evidence has associated the outcome of adult allogeneic hematopoietic stem cell transplantation (HSCT) with the gut microbiome (GM). There is general agreement in the adult transplant community in considering many clinical variables, namely graft-versus-host diseases (GvHD), infections, mortality, and relapse, differently associated with loss of GM diversity and dominance by individual taxa.¹ This has been made possible by the revolution that next-generation sequencing has provided to our ability to characterize the complex microbial communities hosted by our organism. This has greatly improved our comprehension of the relationship between GM, immunity,² and HSCT-related complications. Furthermore, novel and

intriguing approaches to modulate GM have been explored, making it an attractive target for therapeutic interventions.³ For children, HSCT is the mainstay of treatment of several hematological, oncological, and metabolic diseases. Although transplant-related mortality could generally be considered lower when compared to adults, HSCT complications, such as GvHD and bloodstream infections (BSI), affect a high proportion of patients.^{4–6} While there is considerable evidence about this topic in adult HSCT, less is available in the pediatric setting. This particular population offers interesting challenges such as dealing with a rapidly and continuously evolving GM community and the strong impact of complications in growing subjects. These distinct features need to be addressed in future experimental approaches aimed at exploring GM dynamics in transplanted children. This review aims to provide a critical and comprehensive overview of the state of the art of GM research in pediatric patients receiving HSCT and GM-based therapeutic approaches.

Abbreviations: aGvHD, acute graft-versus-host disease; BSI, bloodstream infections; FMT, fecal microbiota transplantation; GM, gut microbiome; GvHD, graft-versus-host diseases; HSCT, hematopoietic stem cell transplantation; SCFA, short-chain fatty acids

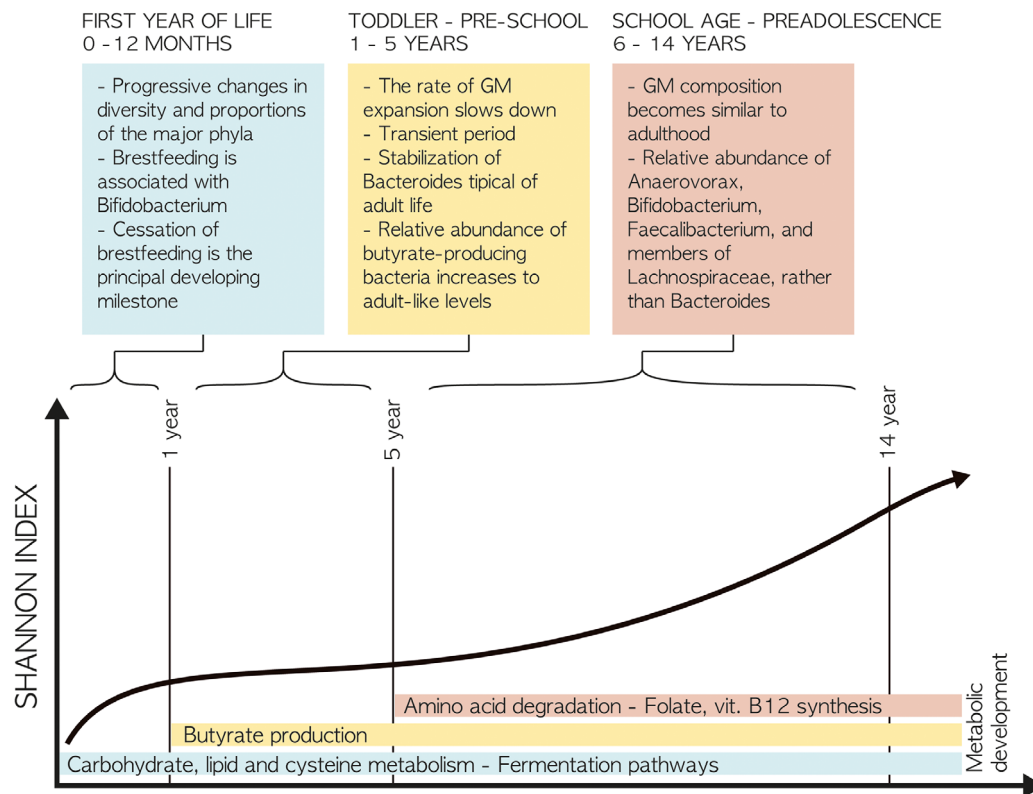


FIGURE 1 Gut microbiome development from birth to preadolescence. The main compositional and functional characteristics of the GM are reported for each of the three development phases: first year of life; toddler and preschool period; school age and preadolescence. Shannon index is a common measure of microbial diversity

2 | GUT MICROBIOME DEVELOPMENT

As our lifelong symbiotic partner, the GM develops along with human aging, featuring hallmark characteristics in different phases of life, and providing “ecosystem services” that seem to respond to the precise physiological needs of each of them. The diversity and structure, both compositional and functional, of the human GM are undoubtedly correlated with age.^{7–13} According to recent and controversial findings, the establishment of the microbiota could begin from pregnancy, driven by the in utero environment.¹⁴ The presence of diverse microbial communities has been described in placenta,¹⁵ amniotic fluid,¹⁶ umbilical cord blood,¹⁷ and meconium,^{18,19} niches that have long been considered sterile. In this section, this review being aimed at describing the impact of the GM in children receiving HSCT, we will focus on the development of GM from birth to adulthood. Although GM development should be imagined as a *continuum*, we can notice some specific features of three main periods, namely: the first year of life, the toddler and preschool period, and the school-age and preadolescence (Figure 1). During the first year of life, consecutive phases of development of the GM have been identified, with progressive changes in diversity and the proportions of the major phyla, up to the establishment of an adult-type profile, with high levels of Firmicutes and codominance of Bacteroidetes.¹² The GM composition at birth is strongly associated with the composition of the mother’s microbiota and the delivery mode.^{12,20–22} However, these differences are short-lived, since they can no longer

be found at 6 weeks of age.⁸ The type of feeding is a well-recognized major factor shaping the infant GM. Breastfeeding is associated with higher levels of *Bifidobacterium*, and the cessation of breast milk represents the principal milestone in the development of the first year of life, driving the shift toward the adult-like structure.¹⁰ During the first postnatal year, progressive changes occur also in the metabolic pathways of the GM, mainly in those involved in carbohydrate fermentation, lipid metabolism,¹⁰ and cysteine metabolism.²³ In the following years, up to the preschool period, the expansion rate of microbiota diversity slows down and the GM shows relative stability. However, there are still some fluctuations that make it difficult to determine when exactly the adult-type profile is established.^{7,10,24} This is thus a transient stage, during which the gradual stabilization of some bacterial communities typical of adult life, such as members of Bacteroidetes, is observed.⁷ It is also worth noting that the relative abundance of butyrate-producing bacteria increases, reaching the adult level at the age of about 5 years.⁷ Butyrate is one of the short-chain fatty acids (SCFA), that are bacterial fermentation products of complex polysaccharides, with a key multifactorial role in host physiology, from a trophic effect on the gut epithelium to protective immunomodulatory capacities.²⁵ Finally, in the school-age and preadolescence period, the GM composition becomes similar to that of adulthood, while still showing some differences in the composition and metabolic profile. Children generally show a higher proportion of *Bifidobacterium*, *Anaerovorax*, *Faecalibacterium*, and members of the *Lachnospiraceae* family, rather than *Bacteroides*,

as described for adults.¹¹ Furthermore, in preadolescence, gut bacterial communities are enriched in growth-supporting functions, such as amino acid degradation, oxidative phosphorylation, and folate and vitamin biosynthesis, in particular vitamin B12.¹¹ Interestingly, the relative abundance of microbial genes involved in steroid hormone biosynthesis increases only during the preadolescent period.²⁶

Clinicians and researchers interested in studying the GM during HSCT should, therefore, take into account the age of the children and their GM profile. For example, as we have described, the GM composition of 1-year-old or younger patients is distinct from that of older children.²⁷

3 | GUT MICROBIOME TRAJECTORY ACROSS ALLOGENEIC HSCT AND CLINICAL OUTCOMES

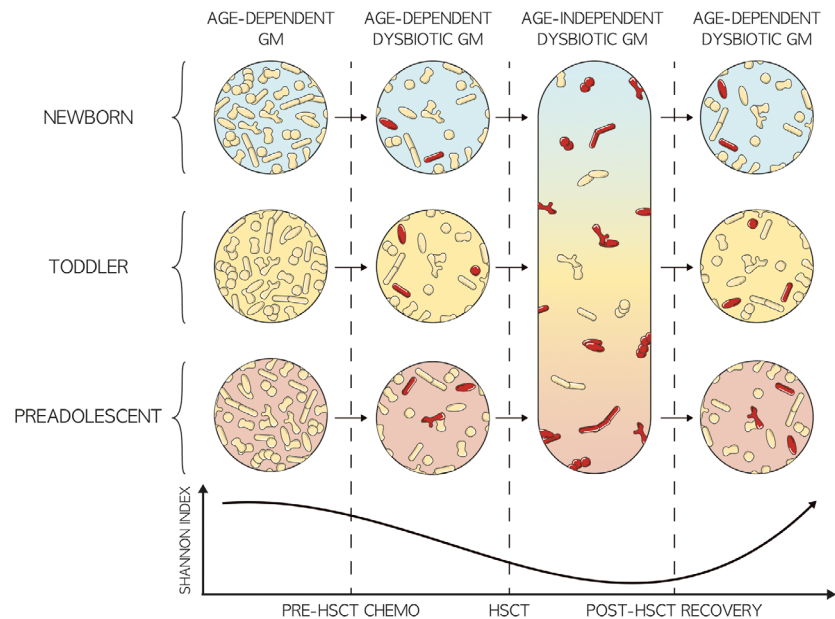
During HSCT, a combination of upsetting events such as the conditioning regimen, exposure to antibiotics, diet, and antacid prophylaxis greatly affects the GM structure, leading to the disruption of its mutualistic asset, with the establishment of the so-called dysbiosis. In pediatric patients, longitudinal examination of the GM in 10 pediatric patients undergoing HSCT for hematological disorders conditioned with busulfan has revealed the loss of about 30% of pre-HSCT alpha diversity, accompanied by a reduction in the proportions of health-promoting taxa, such as *Faecalibacterium* and *Ruminococcus*.²⁸ This paper, together with another study on five pediatric patients undergoing HSCT for acute lymphoblastic leukemia conditioned with total body irradiation, revealed also a massive “invasion” of allochthonous microbes, with only 10% of the preexisting taxa being preserved and greater colonization by genera uncommonly found in healthy subjects, such as *Enterococcus*, *Staphylococcus*, and *Enterobacter*.²⁹ The disruption of the intestinal microbial ecosystem affects not only its composition but also its functionality. This can be documented by alteration of the GM metabolome, as demonstrated by the 76% reduction in the fecal amount of SCFA, with propionate as the most decreased.²⁸ Approximately 2 months after the day of the infusion of the stem cells, in most children the ecosystem recovers the initial richness and metabolic capability; in some cases, this process can take up to 5 months or even longer.^{28,29} Interestingly, a similar trend of disruption and recovery seems to be observed in the oral microbiota,³⁰ suggesting that the latter may provide valuable insights into the perturbation of the GM, especially in children in whom the collection of fecal samples, during such a highly sensitive period of illness, can be very challenging. Even if the GM composition before HSCT is age-related, as mentioned above, in the samples taken at engraftment these differences are no longer detectable, highlighting the extensiveness of the perturbation induced by HSCT that temporarily cancels the age-dependent signature.²⁷ Later in the transplantation course, the GM signature recovers in an age-dependent manner. As briefly discussed, in adults these extreme shifts and stress on GM have been related with HSCT outcomes. A number of excellent studies have associated particular groups of bacteria in the intestinal flora with relapse/progression of disease after allo-HSCT³¹ and higher GM diversity at engraftment with

lower mortality.¹ Data are missing in the pediatric setting, underlining the need for extensive in-depth studies with pediatric patients to confirm the association between GM composition and mortality. In this regard, a small study has shown a higher number of deaths in children with lower bacterial diversity pre-HSCT, even though the number of patients included was too low to draw any definitive conclusions.³² Also, Rashidi et al observed that pre-HSCT gut colonization by *Enterococcus casseliflavus*, intrinsically resistant to vancomycin, is associated with reduced post-HSCT mortality in a mixed cohort of adult and pediatric patients, but no specifics on any age-related differences of GM were provided.^{33,34}

3.1 | Acute graft-versus-host disease

Pediatric patients have a lower risk of developing acute GvHD (aGvHD) than adults; however, this risk is still significant and the associated long-term sequelae of this complication and its treatment may be dramatic in children due to the possible impact on growth.^{35,36} The complex interplay between intestinal mucosa damage, dendritic cells, and alloreactive T cells is influenced by particular variations of the GM architecture after HSCT, which can either promote or protect against aGvHD.³⁷ In pediatric patients who develop aGvHD, several changes occur after allogeneic HSCT, such as: (a) depletion of commensal anaerobes, particularly those belonging to the order Clostridiales, including the well-known health-promoting butyrate producer *Faecalibacterium*; (b) a high percentage of *Enterococcus*; and (c) an invasion of the ecosystem by opportunistic bacteria.^{28,38} Moreover, before HSCT a dysbiotic GM signature predictive of subsequent grade II-IV gut aGvHD can be observed in children, as we found in a multicenter study comprising 36 pediatric patients from four Italian transplantation centers conditioned with myeloablative busulfan or treosulfan-based regimens.²⁷ This signature was characterized by reduced diversity, lower *Blautia* content and increased relative abundance of *Fusobacterium*.²⁷ In particular, *Blautia* is a symbiotic acetate-producing genus of Clostridiales with important antiinflammatory properties, which has been strongly associated in adults with reduced lethal aGvHD and improved overall survival.³⁹ The distinctive trajectory of post-HSCT GM reconstruction in aGvHD subjects is therefore extremely likely to be influenced by the pre-HSCT GM signature. This dysbiotic GM layout might have more chances during HSCT to evolve into a “pathobiome” that is unable to support immunological recovery and promotes aGvHD and other complications. The microbiome changes from before to immediately after HSCT are highly individual-specific in children and, on average, smaller and with greater interpatient variation in the case of higher grade gut aGvHD.²⁷ This may imply that something has already been compromised in the GM before HSCT, and the higher interindividual variability could be related to what has recently been called the “Anna Karenina principle,” stating that each dysbiotic microbial ecosystem is dysbiotic in its own way⁴⁰ (Figure 2). Specifically, this principle refers to dispersive or volatile dysbiosis, as frequently observed for disorders with a strong immunological basis, whose final outcome is determined in part by chance effects such as exposure. The correlation found in

FIGURE 2 Gut microbiome (GM) trajectory during hematopoietic stem cell transplantation (HSCT). The pre-HSCT signature is age dependent. Chemotherapy and other treatments occurring pre-HSCT may induce GM dysbiosis, but the overall layout maintains its age dependence. The magnitude of HSCT-induced dysbiosis causes the GM to lose its age signature with high interpatient variability. This is expression of the “Anna Karenina principle” stating that each dysbiotic microbial ecosystem is dysbiotic in its own way.⁴⁰ During post-HSCT reconstitution, GM recovery is age dependent, sometimes with dysbiotic features promoting complications



children between specific microbial taxa and aGvHD is consistent with that observed in adults. This is especially true for the protective effect of Clostridiales,⁴¹ but the specific role of the various families and genera belonging to this order needs to be better addressed. The role of *Enterococcus* spp. is instead still debated in pediatric patients. While most studies are consistent in linking the expansion of *Enterococcus* with adverse outcomes, mainly aGvHD,^{27,28,32,38} in agreement with the findings in adults,⁴² others have not clearly found this association,⁴³ stressing the need for analysis resolution at species if not strain level. The bacterial family Lactobacillaceae is another prominent member of children's GM that might also have a distinctive role yet to be fully understood.^{32,38,43} Age-related changes in the GM must then be taken into account when investigating the relationship between GM and aGvHD. For example, some studies have observed a protective effect of the Bacteroidetes phylum, in particular the genera *Parabacteroides* and *Bacteroides*,^{28,38} not confirmed in other experiences.^{27,43} This association is mainly found in the pediatric population and may be a signature of the different architecture of GM in children. The ever-changing gut microbial community across different ages of infancy perhaps hinders the discovery of an exact correlation.

3.2 | Immune reconstitution

The close interaction and interlink between immune system reconstitution, specific GM dynamics and outcome after HSCT was thoroughly examined by Ingham et al in 2019.⁴³ They performed a single-center study monitoring 37 patients with malignant and nonmalignant diseases conditioned with myeloablative busulfan, treosulfan, or total body irradiation-based regimens. Children with rapid NK and B cell reconstitution, defined as higher cell counts in the first and second month from allo-HSCT, exhibited high proportions of obligate anaerobic bacteria belonging to the Clostridiales order, specifically to the

Lachnospiraceae and Ruminococcaceae families, including well-known SCFA producers. This more favorable immunological and microbiological pattern was associated with lower aGvHD severity and mortality (Table 1). Patients with higher relative abundances of facultative anaerobic bacteria (Enterobacteriaceae, *Staphylococcus* spp., and *Streptococcus* spp.) showed instead slow NK and B cell reconstitution and higher levels of C-reactive protein. These patients were younger compared to the rest of the cohort, and the authors speculated that an immature GM might exhibit higher susceptibility to potential opportunistic pathogens. No correlation was found between GM and T cell reconstitution. Patients with moderate to severe aGvHD and higher mortality rates showed increased Lactobacillaceae amounts and higher plasma concentration of the antimicrobial peptide human beta-defensin 2 produced by macrophages, prior and after HSCT.⁴³ Systemic inflammation and innate immune system activation may be explained by translocating pathogen- and microbiome-associated molecular patterns (PAMP and MAMP), which can interact with Toll-like receptors and induce the release of inflammatory cytokines.

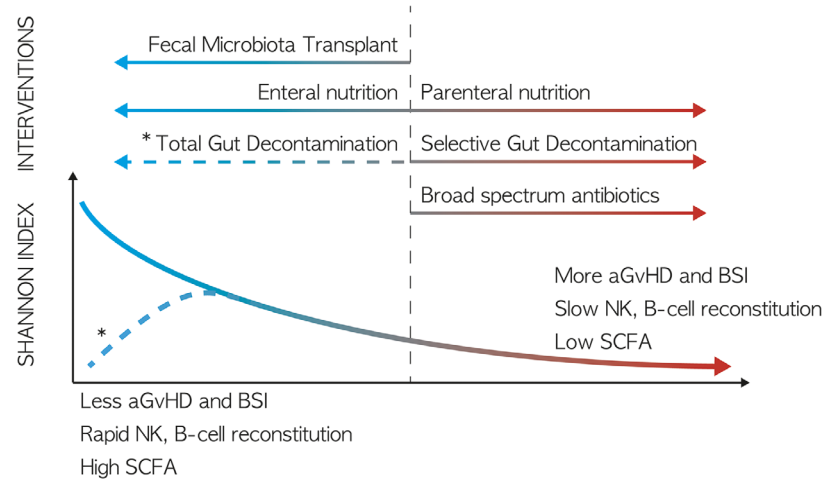
3.3 | Bloodstream infections derived from the gut microbiome

The relationship between infections after HSCT and GM requires a deeper understanding of the pediatric setting, considering the different epidemiology and management of infections in children⁴⁴ and the lack of studies regarding this topic. Bacterial BSI are an important and feared complication of allogeneic HSCT. The gastrointestinal tract is the source of most of these infections, as a result of the injury to the mucosal surfaces caused by the conditioning regimen and impaired nutrition, facilitating the translocation of bacteria into the bloodstream. The only pediatric study on the matter observed that an increase in relative abundances of bacterial strains in the GM

TABLE 1 Main bacterial taxa associated with specific outcomes in pediatric HSCT

PhylumClassOrder FamilyGenus	Patients	Methods of gut microbiome (GM) analysis	Association with hematopoietic stem cell transplantation (HSCT)	Reference
Firmicutes Clostridia Clostridiales Lachnospiraceae <i>Blautia</i>	36 Pediatric patients from 4 centers	Fecal samples collected at three time points: before HSCT, at the time of engraftment, and >30 days following HSCT Analysis of the V3-V4 region of the 16S rRNA gene	Symbiotic acetate-producing genus with important antiinflammatory properties; protective effect toward GvHD	Biagi et al, 2019 ²⁷
Firmicutes Clostridia Clostridiales Lachnospiraceae Ruminococcaceae Any	37 Pediatric patients from 1 center	Weekly fecal samples collection during the first 6 weeks after transplantation Analysis of the V4-V5 region of the 16S rRNA gene	Associated with rapid NK and B cell reconstitution after HSCT, lower aGvHD severity, and mortality	Ingham et al, 2019 ⁴³
	15 Pediatric patients from 2 centers	Weekly fecal samples collection (pre/during/post-HSCT) Analysis of the V4 region of the 16S rRNA gene		Simms-Waldrup et al, 2017 ³⁸
Firmicutes Bacilli Lactobacillales Enterococcaceae <i>Enterococcus</i> (<i>E. casseliflavus</i>)	66 Patients (adults and children) from 1 center	Retrospective analysis of HSCT patients with >1 positive rectal swab or fecal culture for iVRE between days –14 and +14	Associated with reduced post-HSCT mortality in a mixed cohort of adult and pediatric patients	Rashidi et al, 2019 ³⁴
Fusobacteria Fusobacteriia Fusobacteriales Fusobacteriaceae <i>Fusobacterium</i>	36 Pediatric patients from 4 centers	Fecal samples collected at three time points: before HSCT, at the time of engraftment, and >30 days following HSCT Analysis of the V3-V4 region of the 16S rRNA gene	Present in the pre-HSCT signature of patients who develop gut aGvHD grade II-IV	Biagi et al, 2019 ²⁷
Firmicutes Bacilli Lactobacillales Lactobacillaceae Any	37 Pediatric patients from 1 center	Weekly fecal samples collection during the first 6 weeks after transplantation Analysis of the V4-V5 region of the 16S rRNA gene	Present in patients with moderate to severe aGvHD and higher mortality rate	Ingham et al, 2019 ⁴³
Bacteroidetes Bacteroidia Bacteroidales Porphyromonadaceae <i>Parabacteroides</i> <i>Bacteroides</i>	10 Pediatric patients from 1 center	Serial fecal samples collection (pre/during/post-HSCT) Analysis of the V4 region of the 16S rRNA gene	Associated with lower incidence of aGvHD; however, data are discordant ^{27,43}	Biagi et al, 2015 ²⁸
	15 Pediatric patients from 2 centers	Weekly fecal samples collection (pre/during/post-HSCT) Analysis of the V4 region of the 16S rRNA gene		Simms-Waldrup et al, 2017 ³⁸
	5 Pediatric patients from 1 center	Fecal samples collected once during or before conditioning and six times post-HSCT Analysis of the V4-V6 region of the 16S rRNA gene		Lähteenmäki et al, 2017 ²⁹

FIGURE 3 Effect of different types of intervention on the gut microbiome (GM) diversity expressed using the Shannon index



precedes most BSI episodes caused by the same species.⁴⁵ In some cases, this increase results in intestinal dominance that is defined as a single bacterial taxon comprising 30% or more of the GM. Even if this phenomenon is mostly observed in microorganisms that typically have an ecological niche within the lower gastrointestinal tract, sometimes the gut can also serve as a reservoir for typically nonenteric pathogens in HSCT recipients.⁴⁵ In this context, antibiotic-resistant pathogens represent a rising challenge because of the high mortality rates (>50%) observed in pediatric HSCT.^{46,47} The GM has recently been considered as a potential reservoir of antimicrobial resistance. We have addressed this issue in pediatric transplant recipients, focusing on the gut resistome, defined as the overall pattern of antimicrobial resistance genes present in the GM. In children who went on to develop aGvHD, we observed a considerable expansion of the gut resistome in the post-HSCT. Interestingly, this expansion involved not only the consolidation of resistances already present before HSCT but also the acquisition of new genes conferring resistance not limited to the antibiotics administered along the therapeutic course.⁴⁸ This may be critical in guiding the choice of the correct anti-infective treatment in this setting.

4 | MODULATION OF THE INTESTINAL MICROBIOME DURING HSCT

The strong impact of GM dysbiosis on allogeneic HSCT complications has led to the fascinating idea of modulating the GM structure in order to improve clinical outcomes. Several methods exist to interact with the GM of children (Figure 3).

4.1 | Gut decontamination and antibiotic prophylaxis: A change of view

The improved understanding of the microbiota has brought a change of view of the GM as a risk factor for the cited HSCT complications. As suggested by Pitlik, we should not frame complications in the context of Koch's postulate, but rather as microbial individuals interacting in a social fashion.⁴⁹ Gut decontamination and antibiotic prophylaxis

can serve as an example of this change of view. Starting from some studies on a mouse model in the 1970s, some authors hypothesized that total eradication of the intestinal flora could protect patients from infections and aGvHD.^{50,51} The transition from the mouse model to human patients was harder than expected, since there were some inconsistencies both in terms of results and in terms of efficacy of the procedure.^{44,52–57} Furthermore, recent studies have demonstrated that effective antibiotics against anaerobic bacteria after HSCT, such as vancomycin and ciprofloxacin, have detrimental effects on the GM. In particular, a multicenter study analyzing 15 pediatric patients undergoing HSCT found that this type of treatment is associated with the depletion of commensal Clostridiales anaerobes, particularly Ruminococcaceae and Lachnospiraceae members. This GM disruption was associated with the development of aGvHD, demonstrating a detrimental effect also on clinical outcomes.³⁸ Another study on the GM of 42 pediatric patients receiving antibiotics with activity against anaerobic organisms showed an increase in the proportions of facultative anaerobic bacteria as Enterobacteriaceae, and with lower butyrate and propionate levels at engraftment,⁵⁸ especially in children who develop aGvHD.^{38,58} The use of ceftazidime instead showed a positive correlation with high Clostridiales levels.⁴³ Bekker et al analyzed the effect of selective and total gut decontamination in 19 patients undergoing allogeneic SCT for malignancy, primary immunodeficiency, myelodysplastic syndrome, and hemoglobinopathy. The authors evaluated in an ecological fashion the microbial alterations, pointing out the microbial dysbiosis consequent to the massive use of antibiotics.⁵⁹ This should be carefully considered by clinicians who must take into account at the same time the pathogenic potential of bacteria and the risk of antibiotic-mediated dysbiosis. Thus, when trying to interact with the GM, we should always consider the impact of any intervention on the whole ecosystem. In light of this, here we propose the Delphic boat paradigm, as opposed to Koch's postulate. The geneticist Antoine Danchin once used the parable of Delphic boat to metaphorically describe the complexity of the genes, and we can borrow the same image to describe the microbiota interactions.⁶⁰ In the parable, the oracle of Delphi is asked to consider a boat: if, in time, every plank has rotted and been replaced, is the boat the same boat? The answer is that the boat is the same, because it is not its planks but the relationship

TABLE 2 Clinical trials on the use of probiotics and FMT in pediatric HSCT

Treatment	Trial number	Design	Conditions	Status
Oral <i>Lactobacillus plantarum</i>	NCT01010867	Single group assignment Multicenter	Intestinal dysbiosis in patients undergoing hematopoietic stem cell transplantation (HSCT)	Completed
<i>Lactobacillus</i> CD2 lozenges	NCT01480011	Randomized Single center	Oral mucositis in patients undergoing HSCT	Completed
Oral <i>Lactobacillus plantarum</i>	NCT03057054	Randomized Multicenter	Malignant neoplasm in patients undergoing HSCT	Recruiting
Fecal microbiota transplantation (FMT)	NCT03549676	Single group assignment Single center	Refractory graft-versus-host disease (GvHD)	Not yet recruiting
FMT with ruxolitinib	NCT04269850	Single group assignment Single center	Severe acute intestinal GvHD	Recruiting
FMT	NCT03148743	Single group assignment Single center	Gut acute GvHD treated	Recruiting

among them. Similarly, we should face the GM as a complex interacting ecosystem and not only as the raw sum of its several different microbial species.

4.2 | Could nutrition be an effective way of modulating the GM?

The connection between nutrition and human microbiome in maintaining human health is well established.^{61,62} A diet richer in fiber in children during the pre-HSCT period seems to correlate with earlier neutrophil engraftment and a shorter duration of febrile neutropenia,⁶³ even if a detailed assessment of the GM in the pre-HSCT period based upon the type of diets has never been performed. The type of nutritional support during HSCT may also be a key factor in maintaining GM eubiosis. The use of enteral nutrition instead of parenteral nutrition could aid the recovery of GM homeostasis, thus minimizing dysbiotic shifts and downstream impact on patient health. In particular, prompt recovery of a diverse GM layout and SCFA production profiles post-HSCT was seen in enterally fed children, while the GM of parenteral nutrition recipients failed to recover the eubiotic structure. Moreover, some genera restored in the enteral nutrition group during the post-HSCT recovery were identified, namely *Faecalibacterium*, *Dorea*, *Blautia*, *Bacteroides*, *Parabacteroides*, and *Oscillospira*, which are all well-known health-associated GM components capable of producing SCFA.⁶⁴ These findings are in line with the clinical benefit observed with enteral nutrition regarding aGvHD and BSI.^{65,66} The role of specific nutritional supplements aimed at promoting a GM mutualistic layout, for example, discrete fiber structures to boost directed SCFA production,⁶⁷ has not been explored yet in children undergoing HSCT, but has shown promising preliminary results in adults.⁶⁸ Nutritional strategies, either by changing the route of nutritional supplementation or by administering specific molecules, are thus promising, cost- and risk-effective methods of modulating the GM, worthy of being further explored in children.

4.3 | The challenge of fecal microbiota transplantation in children

Historically, the GM can also be modulated using live microorganisms or microbial consortia, from traditional probiotics to next-generation probiotics (also referred to as live biotherapeutics⁶⁹), up to the most recent fecal microbiota transplantation (FMT). FMT consists of the infusion of fecal matter from a healthy donor into the gastrointestinal tract of a recipient patient harboring a dysbiotic GM. This procedure directly modifies the host's GM composition in an attempt to restore eubiosis and gut homeostasis. In adult patients, it has been used to treat refractory *Clostridium difficile* infections and steroid-resistant gut aGvHD, and to eradicate drug-resistant microorganisms.^{70–72} Blue-stone et al reported FMT administration to treat refractory *C. difficile* infection in three children, using related and unrelated donors via a gastric tube or colonoscopy. No adverse events were reported, but only one achieved remission.⁷³ A retrospective study focused on adult hematologic patients reported a 16-year-old girl receiving two FMT for vancomycin-resistant enterococci and carbapenemase-producing bacteria colonization after allogeneic HSCT, resulting in the decolonization of the former and persistence of the latter, with no reported adverse events.⁷⁴ Merli et al performed FMT in five pediatric patients before allogeneic HSCT using samples from the same donor, to induce decolonization of multidrug-resistant bacteria in the gut. The procedure had a good safety profile; however, long-term colonization was not achieved in four out of five patients.⁷⁵ Several practical and microbiological issues arise when contemplating FMT in pediatric allogeneic HSCT, and they need to be addressed in the near future to optimize its use in clinical practice. Exact indications of FMT, source of fecal material, donor screening, route of administration, use of antibiotics, and safety profile are the most glaring issues to be examined. For instance, the safety of the administration of live microorganisms to immunocompromised children with impaired gut permeability is strongly debated. While most reports deem this practice safe,⁷⁶ there are still concerns that need to be addressed^{77,78} (Table 2).

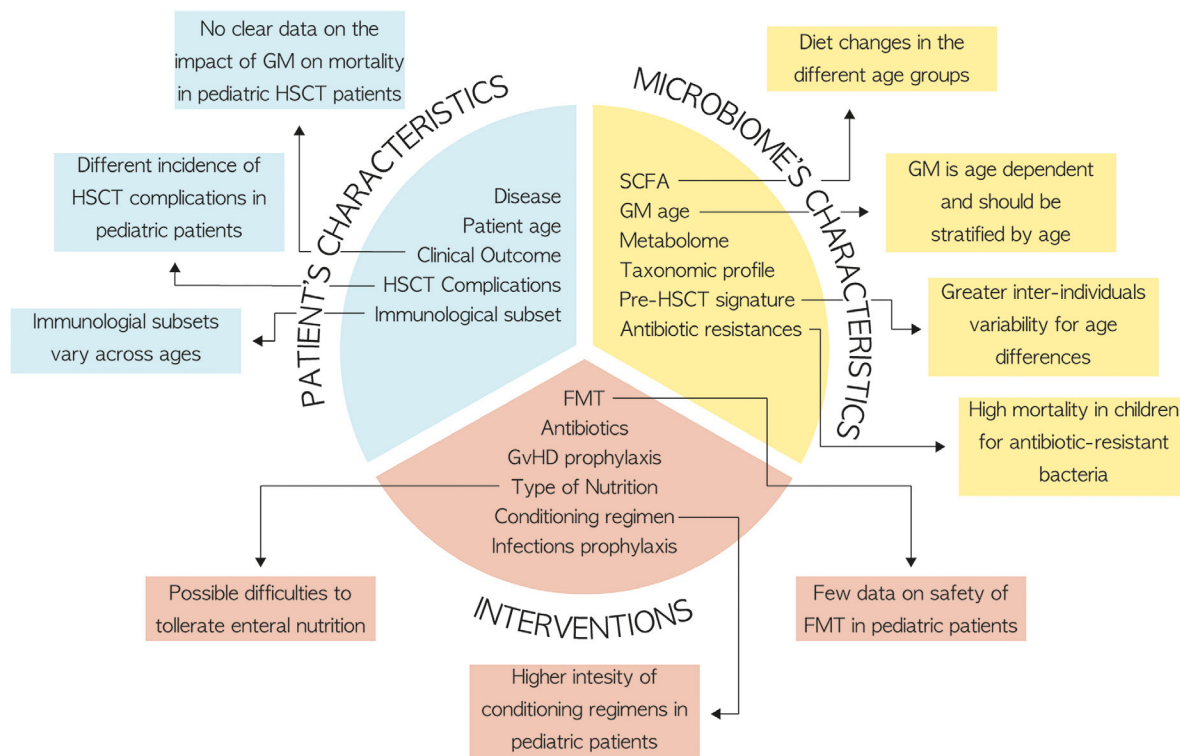


FIGURE 4 Structuring the design of exclusive pediatric studies: variables to take into account and challenges to face

5 | CONCLUSIONS AND FUTURE DIRECTIONS

The evidence reported in this review highlights the increasing knowledge of the role of GM in allogeneic HSCT pediatric recipients, but numerous unsolved questions are yet to be answered. Children have specific characteristics in terms of both indications for transplantation and complications.⁴⁴ Moreover, GM is age-related and develops in tight connection with a developing immune system in an age-dependent manner.⁷⁹ To date, no experience in the literature has thoroughly examined the impact of patients' age on GM modifications. These considerations stress the need for studies in exclusively pediatric cohorts to evaluate whether the results obtained in adults on the impact of GM dysbiosis on mortality¹ and transplant-related complications are generalizable to pediatric cohorts. Moreover, studies exclusively enrolling children should take into account the unique pediatric-specific variables and face related challenges (Figure 4). Among these, the heterogeneous pre-HSCT signature and the different response to the upsetting events occurring during transplantation of an age-dependent GM interlinked with a developing immune system must be a key topic of research in the near future. In conclusion, the field of research on microbiota and allogeneic HSCT has grown exponentially in the last decade, potentially with relevant implications for therapeutic interventions. A focus on the particular features of children in the coming years is awaited in order to further improve clinical outcomes in these patients.

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest.

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